

DEBRE BERHAN UNIVERSITY

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Master Thesis

**PHYTOCHEMICAL INVESTIGATION AND ANTI
BACTERIAL ACTIVITY TEST OF THE LATEX EXTRACT OF
EUPHORBIA ABYSSINICA “KULKUAL”**

A Thesis Submitted to the Office of Graduate Program Of Debre Berhan University in
Partial Fulfillment of the Requirements for the Degree of
Master of Science in Chemistry

BY

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**February, 2021 G.C
Debre Berhan, Ethiopia**

DEBRE BERHAN UNIVERSITY
COLLEGE OF NATURAL AND COMPUTATIONAL
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APPROVAL PAGE

A thesis submitted to the office of graduate program of Debre Berhan University in partial fulfillment of the requirements for the Degree of Master of Science in Chemistry.

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As a member of the board of examiners of Master of Science thesis defense examination, we have read and evaluated the thesis work done By Abreham Bantie entitled phytochemical investigation and anti-bacterial activity test of the latex extract of Euphorbia Abyssinica

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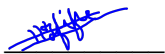
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DECLARATION

I assert that this dissertation entitled “phytochemical investigation and anti-bacterial activity test of the latex extract of *Euphorbia Abyssinica*” is my original work.

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.....

Date

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ABSTRACT

Euphorbia Abyssinica or local name “kulkual” is belongs to the family euphorbiaceae. It grows in moist mountain forest, humid wood lands and scrub savannah up to 10 meters height. At intervals along the stem angles or ribs are small, rounded structures from which leaves, flowers or spines may grow. In succulent euphorbias the spines are paired, whereas in cacti spines are either singular or held in groups of three or more. The species has developed the capacity to store water in its stem tissue. It has been traditionally used for treating a wound, to treating poisons of a mouse, to medicament of rabies and to wart extinction. General phytochemical investigation of the latex parts of *euphorbia abyssinica* done based on petroleum ether (PE), ethyl acetate (EtOAc) and ethanol (EtOH) solvent extracts. The aims of this study was phytochemical investigation of the latex based on different solvent extracts of the plant and examining its anti-bacterial activity. Partitioning each extract was concentrated and the solvents were separated by using rotary evaporator giving the following yields petroleum ether extract 3.40%, ethyl acetate extract 6.98%, and ethanol extract 3.187%. The phytochemical tests of the petroleum ether extract of the latex part display the presence of saponins, steroids, tannins and terpenoids. Flavonoids, saponins, steroids, tannins and terpenoids were observed in ethyl acetate extract. The ethanol extract show the presence of flavonoids, steroids, tannins and terpenoids. Anti-bacterial activity of the crude extract tested on gram positive *Se*, and gram negative *Ec*, and *Sh* bacterial and also *Ca* yeast were available in laboratory show inhibition. And also the fresh latex of the sample show on *Ec* and *Sh* bacterial with inhibition zone 9 mm and 7.

Key words: *Euphorbia abyssinica*, phytochemical, crude extract, anti-bacterial, Soxhlet extractor.

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ACRONYMS

PE	Petroleum ether
EtOH	Ethanol
EtOAc	Ethyl Acetate
DMSO	Dimethyl sulphoxide
WHO	World Health Organization
Ca	Candida alibi cans
Ec	Escherichia coli
Lm	Listeria mono cytogenes
Pa	Pseudomonas aeruginosa
Sa	Staphylococcus aureus
Sal	Salmonella thphimurium
Se	Staphylococcus epidermidis
Sh	Shigella sonnei
Sp	Streptococcus pyogenes

CHAPTER ONE

INTRODUCTION

1.1. Background of the Study

Plants have a wide range of scientific and economic value. They are considerably valuable in agricultural, food, cosmetics, medical, and pharmaceutical industries. They are an essential source of bioactive compounds that can be used to cure or treat diseases. Medicinal plants are an important element of the system in Ethiopia as well as in other countries [1].

Plant extracts, as a pure compounds or standardized extracts are natural products delivering an important chance for new drug invention and increases the accessibility of biochemical diversity. Traditionally available medical plants have different chemical constituents that are used to treat diseases in society and are accepted as a traditional drug to protect and cure sicknesses [2]. Most of the world population trusted to use traditional medicine (ethnopharmacognosy) for their vital healthcare and to cure chronic as well as infectious diseases. Mostly societies, believe that certain herbal products are active and safe with important biological activity such as anticancer, antimicrobial, antioxidant, antidiarrheal, painkilling and wound healing action. The Scientific examination helps to evaluate carefully pharmacokinetics, bioavailability, effectiveness, toxicology and both immediate and long-term side effects of bioactive extracts of the plant before application [3].

The history of plant-based health, care system goes back to ancient times and as old as human civilization. Ancient people use medical plants as simple curative forms by mixing different plant parts, by fermenting and by macerating numerous plant parts. Different parts of plants have been the primary source of medicines in the traditional health care systems around the world, recently and even currently in most of the developing countries. The approach of characterization and isolation of active ingredients from plants started in the late 19th century. Consequently, chemical substances isolated from plants had currently used for drug production. The newly approved drugs were synthesized from natural products of the plant on the bases of the traditional implications of plants [4].

The therapeutic use of plants and herbs are occasionally investigated while people move around the environment to search for food for the survival of life. They believe that plant, and plant products are effective therapeutic tools for fighting against diseases. The way of controlling diseases with the help of plants that are available as traditional medicinal plants provide an important approach to establish synthetic drugs [5].

Corresponding to synthetic drug demand, the global natural products market is growing exponentially. The demands for chemical diversity in screening programs increase throughout the world due to seeking therapeutic drugs from natural products particularly from traditional medicinal plants and edible plants. Investigation on the medicinal benefit of botanicals and herbals shows the presence of various types of bioactive compounds with therapeutic activity. The medicinal value of those plants depends on bioactive constituents of phytochemicals present in the plant and their action on human body [6].

Numerous plant species are recorded to have active phytochemical compounds which are known as plant secondary metabolites [7]. Secondary metabolites act as a plant's natural system of defense and repair. Two-thirds of the world's plant species have medicinal importance, and almost all of these have excellent antioxidant potential [8].

Phytochemicals or plant secondary metabolites are not essential nutrients and are not required by the human body for sustaining life but have important properties to prevent or to fight some common diseases [9]. They are naturally found in all parts of the plant body bark, leaves, stem, root, flowers, fruits, seed, latex etc. that is any part of the plant body may contain bioactive components [10].

Surveys on therapeutic applications of plants led to the discovery of several clinically valid drugs by characterizing the components (e.g., digoxin, morphine, reserpine, taxon, vinblastine, vincristine etc.). Elucidation and characterization of the structure of active components paved the way for extraction and synthesis of compounds with higher efficacy in healing disease and for lowering their adverse effects on human fitness. The major constituents of primary metabolites are carbohydrates, amino acids, proteins and chlorophyll while secondary metabolites are

alkaloids, saponins, steroids, flavonoids, tannins, glycosides, anthraquinones and terpenoids among others[11].

Euphorbia is a diverse plant genus consisting of more than 2000 species with worldwide distribution, chiefly in subtropical and temperate regions. Some species have thick succulent stems and spiny, closely resembling cacti. They are distinguishably different by their peculiar flower and milky latex that contains irritate [12,13]. The species of euphorbia causing ocular toxicity mostly with *E.royaleana*, *E. lathyris*, *E.trucalli*, *E.trigona*¹⁴, *E.ingens*, *E abyssinica* [15], and *E. Candelabrum* [16,17].

The phytochemical screening that carried out on the species Euphorbia abyssinica indicates the presence of Alkaloids, Flavonoids, Glycosides, Saponins, Steroids, tannins, and Terpenoids. Previous studies result showed that there were limited phytochemical investigations and solvents crude extract analysis on the latex part of *Euphorbia abyssinica*. Therefore, this study is expected to fill the knowledge gap and add important facts about the phytochemical constitute of latex parts of *E.abysinica* and show their antibacterial activity. People who work with euphorbia species should wear protective goggles while handling the plant.

1.2. Statement of the problem

Deferent parts of *euphorbia abyssinica* species use many applications.^{18, 19} For example, Latex part of this species was traditionally used in the treatment of several bacterial and virus caused ailments. For example to treat poisons of a mouse, to cure ulcers, to the medicament of rabies. And the seeds of this species are a good quality foundation of food to berry utilize for birds, the flowers are a magnet for butterflies, bees and other insects, and the woods also uses to create doors, planks, and boats. So these active compounds of *euphorbia abyssinica* species extracts were determined experimentally.

However, the latex part of *euphorbia abyssinica* shows deferent biological activities in traditional remedies. This traditional application not only hindered the standardization and development of this herb, but also made its recognition; acceptance and utilization remain locally restricted. Understanding the chemical site and antibacterial activity of *euphorbia abyssinica* milky juice or latex is essential for their practical application.

Therefore, this study was done to identify the phytochemical constituents and to test anti-bacterial and antifungal activity of the latex extracts of *E.abysynica* on *Candida albicans* yeast and *Escherichia coli*, *Listeria monocytogenes*, *Pseudomonas aeruginosa*, *Staphylococcus aureus*, *Salmonella typhimurium*, *Staphylococcus epidermidis*, *Shigella sonnei* and *Streptococcus* bacterial to check some of its medicinal properties.

1.3. Objective of the study

1.3.1. General objective

The general objective of this study is to identify phytochemicals from crude extracts by using organic solvent extracts and investigate its anti-bacterial activity.

1.3.2. Specific objectives

The specific objectives of this study are:

- To extract the latex by using PE, EtOAc, and EtOH extracts.
- To identify phytochemical constituents from the extracted sample.
- To examine the anti-bacterial activity of the latex extracts.

1.4. Significance of the Study

This study provides evidence about the phytochemical availability in the latex part of *E.abysynica* using phytochemical tests. The societies in Awe and west Gojjam zone use the latex part to treat poisons of mouse, to dry wound, to wart extinction, to paste, to medicament of rabies by drink latex with milk. So, phytochemical identification and characterization of the plant extracts are used as an indication for standardization of bioactive components found in the plant latex. The study examines anti-bacterial potential of the latex extract of *E.abysynica* to show the effectiveness of plants in the treatment of diseases and opens the way to prove the importance of its application as traditional medicine.

1.5. Scope of the study

The traditional medicines of the latex part of *E.abbyssinica* survey was conducted in Amahara region, Awe and West Gojjam zone, especially in Dembecha, Jabitenan, Sekela, Guagusa, and Degadamot woreda's. The survey includes some important villages and some towns. Information was obtained using free-listing, semi-structured and open-ended interviews with traditional healers, community elders, and mothers. More medicinal plants that used to treat diseases associated with bacteria were identified and the study justifies the traditional use of this plant in Gojjam traditional medicine.

CHAPTER TWO

2. LITERATURE REVIEW

2.1. *Euphorbia abyssinica*

2.1.1 Description

The plant *Euphorbia abyssinica*, locally known as “*Kulkual*” in Amharic or “*Adami*” in Oromifa [20]. It is a species of plant in the family Euphorbiaceae and it was first described in 1791, by botanist Johann Friedrich Gmelin [21].

Euphorbia abyssinica is a succulent, spiny tree, towering up to 10 meter in height. It either stands large or grows as a solitary plant in moist Mountain forest, humid woodlands and scrub savannah. The plant is endemic to Eastern Africa including Ethiopia, Eritrea, Somalia, and Sudan. It grows at an altitude of 1400 to 2400 meters above sea level. In Ethiopia, it usually grows in steep rocky hillsides, and sometimes used for live fencing at high altitudes. It performs well in dry, moist and wet Weyna Dega agro climatic zones in Tigray, Gondar, Gojjam, Shwa, Harerge and Sidamo, 1,400–2,400 meter above sea-level, usually above 1,900 meter [22].

The thick, green main trunk, which becomes woody with age, supports many 5-8-angled, branching stems. The species has developed the capacity to store water in its stem tissue and is also able to photosynthesis effectively through the surface tissue of the stem. Leaves form on young growth at the end of the branches but fall in the dry season. At intervals along the stem angles or ribs are small, rounded structures from which leaves, flowers or spines may grow. In succulent euphorbias the spines are paired, whereas in cacti spines are either singular or held in groups of three or more [23].

A distinguishing characteristic of euphorbias are the cyathia. These tiny cup-shaped structures bear the very small, modified male and female flowers. The male flowers of *Euphorbia abyssinica* are reduced to a single stamen and the female flowers consist of a stalked pistil with branched stigmas. The flowers are without petals, but are surrounded by noticeable yellow bracts. Tiny glands producing a nectar reward for would-be pollinators surround the flowers.

The fruit is a capsule up to two centimeters long, which, as it ripens, explodes scattering the seed across the ground [24].

Euphorbia abyssinica. It has a toxic milky sap. If the sap exposed to body, it caused rash, irritant dermatitis and blistering, nausea and vomiting. The sap can also cause temporary or permanent blindness if it comes into contact with the eyes. Many plants in the family of Euphorbiaceae contain euphorbon (a dangerous poison) [25]. For the same reason it is not recommended to grow the tree in the garden where children play. However, it is used for firewood and provides timber for (furniture, roofing, matches, boxes, local tables, wooden saddles [26].



Figure 1. Plant species of *Euphorbia abyssinica*.

2.1.2. Scientific classification:

Kingdom: Plantae

Family: Euphorbiaceae

Genus: Euphorbia

Species: *E.abbyssinica*

Binomial name: *Euphorbia abyssinica* J.F.Gmel., 1791 [27]

Synonyms: E: acurensis.N.E.Br

2.2. Importance of Natural Product

Natural products of plant contain a vast and diverse range of organic compounds important for scientific study [28]. The term “natural product” is secondary metabolites commonly used to express those naturally occurring compounds that are unique to one plant, or common to a small number of closely related plants. They are small molecules that exist in organisms with interesting biological activity and chemical properties. The extracts of natural products are more applicable in the area of agriculture, cosmetic, medicine and food science in early as well as in modern societies. So, it is the major interest in the field of scientific research to understand their usefulness, and divers applications in a society [29].

The chemistry of natural product studies covers extraction of organic compounds, identifying their function, elucidation of structure and synthesis. The compounds of plant extracts that are identified, extracted and purified by scientific techniques are active natural products. Natural products have a function in the organism from which they originate. The frequent survival of a particular organism in the environment by safeguarding them is now considered to have many vital roles as mediators of ecological interactions. The ability to synthesize a range of secondary metabolites, which may repel or attract other organisms, is one way of the survival strategy. Flower color, pollination, preventive properties in plants and chemical defense may explain some of the important activities of secondary metabolites for the survival of an organism [30].

2.3. Modern Herbal Medicine

The use of herbs to treat diseases is almost common among developed societies since they contain a variety of compounds or secondary metabolites. WHO evaluates that 80% of the population of developing countries currently uses herbal medicine for principal health care as complementary or alternative medicine. Pharmaceuticals are prohibitively expensive for most of the world's population compared to ethno-medicinals. Those are gathered from plants and obtained easily with little or no cost. The use and search for drugs and dietary supplements derived from plants accelerated in recent time because they have direct or indirect medicinal or dietary provision. Pharmacologists, microbiologists, botanists, and natural-product chemists are searching the Globe for phytochemicals and deal for treatment of several diseases. Among many active compounds currently isolated from the advanced plants most of them are widely used in the modern synthetic drugs. Modern synthetic drugs show 80% positive correlation with traditional therapeutic activity of the plants from which they are derived. More than one-thirds of the world plant species have medicinal value in the developing countries. Great attention paid to choose the extraction procedure in order to get satisfactory yields of high quality and best-valued products, respecting their safety and medicinal value [31].

Several reviews published regarding to the methods of selecting plants for drug synthesis and extraction of active components. Random selection followed by biological activity screening is one approach to extract bioactive components. In this approach, randomly selected plants subjected to various biological activities and comprehensive screening of major phytochemicals such as more sugar molecules cardenolides /bufadenolides, alkaloids, triterpenes, flavonoids, etc. performed using various analytical and quantitative tools. Chemical analysis and biological activity are attributed to qualify and quantify major phytochemical class of the extract using standard methods [32].

The phytochemical tests are simple to perform, but false-positive and false-negative tests often reduce the results and make difficult to assess the activity of extracts. More importantly, it is frequently impossible to relate one class of phytochemicals to a specific biologic activity; for example, the alkaloids or flavonoids produce a vast array of biological effects that are usually not predictable in advance [33].

2.4. Medicinal use of *E.abysinica* species

In Ethiopia, the sap has been used mixed with butter as a topical treatment for fungal skin infections. It is also used by traditional healers in the treatment of malaria and visceral leishmaniasis. There is no ethno botanical data on the use of *Euphorbia abyssinica* as a traditional medicinal plant outside Ethiopia. It is rather classified as a poisonous plant by the Food and Drug Administration of the United States of America [34]. In Ethiopian traditional medical practices, the plant has been utilized for treatment of various diseases including malaria [35,36].

Its medicinal effects are mainly confined to its latex and root. The mode of preparation of the plant for treatment of malaria by local people and/or herbalists was as followed: the fresh root was collected; it was chopped using stone and woody materials; the chopped plant material was dried under sunlight; the dried product was further powdered and eaten with egg then a lot of milk was drunk [37]. However, the safety, quality and efficacy of *Euphorbia abyssinica* for treatment of malaria have not been verified scientifically. In Ethiopia a stem bark decoction of *Euphorbia abyssinica* is taken to treat gastro-intestinal complaints. The latex is taken in milk or eaten with bread to treat gonorrhoea or ascariasis. It also applied to: wounds to accelerate healing and to ticks of cattle [38].

Although other species that have the same family and genus and also the same structure and synonyms with *E.abysinica* have been reported used as:

2.5.1. *Euphorbia candelabrum*

Occurs from Somalia, Sudan, Ethiopia, Congo, Zimbabwe, Kenya, Tanzania, etc. [39,40]. Its latex is very toxic and may cause blindness when it comes into contact with the eyes. It is also blistering and irritating to the skin and mucous membranes. In Central Africa several drops of latex diluted in water are taken to treat coughs and tuberculosis or as an emetic and abortifacient. The latex is also applied to wounds, sores and warts [41]. The Loita Maasai people mix the latex with fat, which is rubbed on the body to treat malaria. Latex is taken in porridge as a strong purgative, and to treat HIV infection [42].

In Kenya and Tanzania roots are boiled with chicken or meat, or with stomach fluids from a slaughtered goat or cow, and the liquid is drunk as a strong emetic to treat stomach-ache, severe constipation and infertility [43]. Latex is taken in porridge as a strong purgative, and to treat HIV infection. In Kenya the latex is also applied to wounds and sores of cattle. Stem ash is powdered and used to treat eye infections, and it can be used as firewood, although the smoke is irritate. The latex is an ingredient of arrow poison. Fresh, pounded branches are thrown into watering holes and streams as a fish poison and to poison wild animals and the sticky latex is used as birdlime [44].

The light, durable wood is used to make roofing, tables, doors, matches, boxes, mortars, musical instruments and saddles. The trunk split into halves is hollowed and re-joined to make beehives. Cut branches are used as fencing and planted for shade. The flowers of *Euphorbia candelabrum* produce much nectar, but the honey causes a burning sensation in the mouth, which is intensified by drinking water [45,46].

2.5.2. *Euphorbia ingens*

E.ingens occurs in Malawi, Mozambique, Zimbabwe, Zambia, Botswana, Tanzania, South Africa (in the north and east of the country), Rwanda, Uganda, Kenya and Swaziland [47,48].

Flowers of this tree are a magnet for butterflies, bees, and other insects, which gather pollen and nectar from them, pollinating the trees in the progression. Seeds are a good quality foundation of food for many fruit and berry utilize birds. Birds also like nesting in this foliage, whole nesting birds such as woodpecker's often used dead sections.⁴⁹ The latex of this trees immensely toxic and be able to cause severe skin irritation, loss of sight, and severe sickness to human and animals if swallowed [50]. If properly applied, it can be medicinally used as a purgative or for the cure ulcers. It's said to be used by the Venda and Sotho people as a cure for cancer. Kindling is used as fish poison in South Africa and Zimbabwe [51]

2.5.3. *Euphorbia ampliphylla*

Pax (synonym: *Euphorbia obovalifolia* auct. non A.Rich.) is a montane forest tree and occurs from Ethiopia and Somalia south to Malawi and Zambia. In Ethiopia a little latex, probably *in* water, is taken orally to treat rabies. In Kenya a wood decoction is given to treat stomach-ache during childbirth [52,53].

2.6. Phytochemical Examination of the Plant Species

Phytochemical examination of a plant involves extraction of the plant material, separation, isolation of the ingredients of interest, characterization of the compounds and quantitative evaluations. Phytochemical examination is the way of identifying the presence of bioactive constituents in the plant extracts [54].

2.6.1. Alkaloids

Alkaloids are a class of basic, naturally occurring organic compounds that contain at least one nitrogen atom. This group also includes some related compounds with neutral and even weakly acidic properties [55]. Some synthetic compounds of similar structure may also be termed alkaloids [56]. In addition to carbon, hydrogen and nitrogen, alkaloids may also contain oxygen, sulfur and, more rarely, other elements such as chlorine, bromine, and phosphorus. Alkaloids are produced by a large variety of organisms including bacteria, fungi, plants, and animals[57]. They can be purified from crude extracts of these organisms by acid-base extraction, or solvent extractions followed by silica-gel column chromatography. Alkaloids have a wide range of pharmacological activities including antimalarial (e.g. quinine) , antiasthma (e.g. ephedrine), anticancer (e.g. homo-harringtonine), cholinomimetic (e.g. galantamine), vasodilatory (e.g. vincamine) , anti-arrhythmic (e.g. quinidine), analgesic (e.g. morphine) [58], antibacterial (e.g. chelerythrine) [59], and anti-hyperglycemic activities (e.g. piperine), Many have found use in traditional or modern medicine, or as starting points for drug discovery. Other alkaloids possess psychotropic (e.g. psilocin) and stimulant activities (e.g. cocaine, caffeine, nicotine, theobromine and have been used in entheogenic rituals or as recreational drugs [60].

2.6.2. Flavonoid

Flavonoids (or bioflavonoids) (from the Latin word flavus, meaning yellow, their color in nature) are a class of polyphenolic plant and fungus secondary metabolites [61].

Uses of flavonoids

I. Inflammation

Inflammation has been implicated as a possible origin of numerous local and systemic diseases, such as cancer, cardiovascular disorders, diabetes mellitus, and celiac disease. Preliminary studies indicate that flavonoids may affect anti-inflammatory mechanisms via their ability to inhibit reactive oxygen or nitrogen compounds. Flavonoids have also been proposed to inhibit the pro-inflammatory activity of enzymes involved in free radical production, such as cyclooxygenase, lipoxygenase or inducible nitric oxide synthase, and to modify intracellular signaling pathways in immune cells or in brain cells after a stroke [62].

Procyanidins, a class of flavonoids, have been shown in preliminary research to have anti-inflammatory mechanisms including modulation of the arachidonic acid pathway, inhibition of gene transcription, expression and activity of inflammatory enzymes, as well as secretion of anti-inflammatory mediators [63].

II. Cancer

Clinical studies investigating the relationship between flavonoid consumption and cancer prevention/development are conflicting for most types of cancer, probably because most human studies have weak designs, such as a small sample size. Two apparent exceptions are gastric carcinoma and smoking-related cancers. Dietary flavonoid intake is associated with reduced gastric carcinoma risk in women, and reduced aero digestive tract cancer risk in smokers [64].

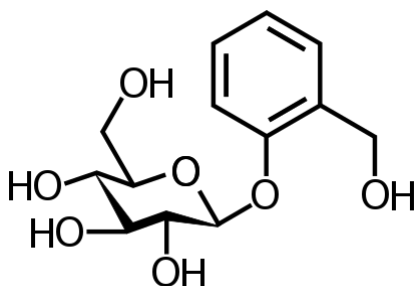
III. Antibacterial

The plants with flavonoids as their major constituents can inhibit *Helicobacter pylori*-infection and use as anti-peptic disease [65].

2.6.3. Glycoside

In chemistry, a glycoside is a molecule in which a sugar is bound to another functional group via a glycosidic bond. It play numerous important roles in living organisms. Many plants store chemicals in the form of inactive glycosides. These can be activated by enzyme hydrolysis which causes the sugar part to be broken off, making the chemical available for use [66].

Many such plant glycosides are used as medications in the form of chemical defense against predators. In animals and humans, poisons are often bound to sugar molecules as part of their elimination from the body [67].



Scheme 2. 1.Salicin a glycoside related to aspirin.

2.6.4. Saponins

Saponins are class of chemical compounds found in a particular abundance in various plant species. More specifically, they are amphipathic glycosides grouped phenolmenologically by the soap-like foam they produce when shaken in aqueous solutions, and structurally by having one or more combined with a lipophilic triterpene or steroid derivative [68,69]

Uses of saponin

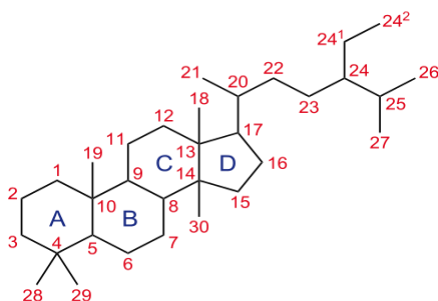
The amphipathic nature of saponins gives them activity as surfactants with potential ability to interact with cell membrane components, such as cholesterol and phospholipids, possibly making saponins useful for development of cosmetics and drugs. While saponins are promoted commercially as dietary supplements and are used in traditional medicine, Saponins are used for their effects on ammonia emissions in animal feeding [70]. In the United States, researchers

are exploring the use of saponins derived from plants to control invasive worm species, including the jumping worm [71].

2.6.5. Steroid

A steroid is a biologically active organic compound with four rings arranged in a specific molecular configuration. They have two principal biological functions: as important components of cell membranes which alter membrane fluidity; and as signaling molecules. Hundreds of steroids are found in plants, animals and fungi. All steroids are manufactured in cells from the sterols lanosterol (opisthokonts) or cyclo-artenol (plants). Lanosterol and cyclo-artenol are derived from the cyclization of the triterpene squalene [72].

The steroid core structure is typically composed of seventeen carbon atoms, bonded in four "fused" rings: three six member cyclo-hexane rings (rings A, B and C in the first illustration) and one five member cyclo-pentane ring (the D ring). Steroids vary by the functional groups attached to this four-ring core and by the oxidation state of the rings. Sterols are forms of steroids with a hydroxy group at position three and a skeleton derived from cholestane [73], and can also be more radically modified, such as by changes to the ring structure. For example, cutting one of the rings. Cutting Ring B produces seco-steroids one of which is vitamin D₃. Examples include the lipid cholesterol, the sex hormones estradiol, testosterone, and the anti-inflammatory dexamethasone [74]



Scheme 2. 2. Structure of cholesterol steroid with 27 carbon atoms.

Its core ring system (ABCD), composed of 17 carbon atoms, is shown with IUPAC-approved ring lettering and atom numbering [75].

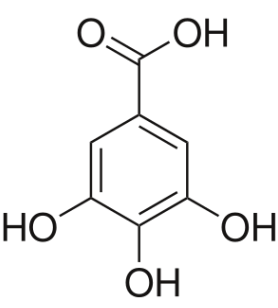
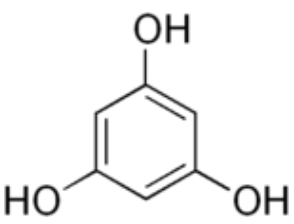
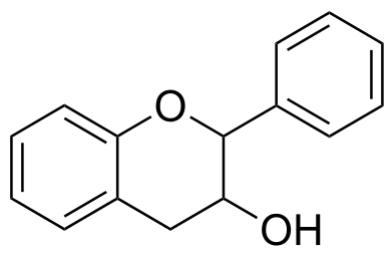
2.6.6. Tannin

Tannins (or tannoids) are a class of astringent, poly-phenolic biomolecules that bind to and precipitate proteins and various other organic compounds including amino acids and alkaloids. The tannin compounds are widely distributed in many species of plants, where they play a role in protection from predation (including as pesticides) and might help in regulating plant growth [76]. The astringency from the tannins is what causes the dry and puckery feeling in the mouth following the consumption of un ripened fruit, red wine or tea [77]. Likewise, the destruction or modification of tannins with time plays an important role when determining harvesting times. Tannins have molecular weights ranging from 500 to over 3,000 (Gallic esters) and up to 20,000 (proanthocyanidins) [78].

Structure and classes of tannins

There are three major classes of tannins: Shown below are the base unit or monomer of the tannin. Particularly in the flavone-derived tannins, the base shown must be (additionally) heavily hydroxylated and polymerized to give the high molecular weight polyphenol motif that characterizes tannins. Typically, tannin molecules require at least 12 hydroxyl groups and at least five phenyl groups to function as protein binders [79]

Table 1. Structure and classes of tannins.

Base Unit:	 Gallic acid	 Phloroglucinol	 Flavan-3-ol's scaffold
Class/Polymer:	Hydrolyzable tannins	Phlorotannins	Isomerized condensed tannins) [80].

Uses

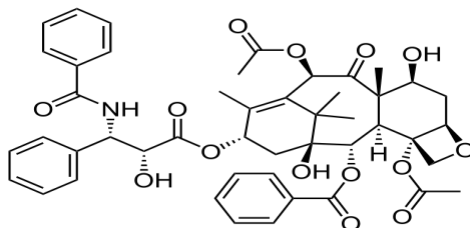
Tannins are an important ingredient in the process of tanning leather. Tanbark from oak, mimosa, chestnut and quebracho tree has traditionally been the primary source of tannery tannin, though inorganic tanning agents are also in use today and account for 90% of the world's leather production [81].

Tannins produce different colors with ferric chloride (either blue, blue-black, or green to greenish-black) according to the type of tannin. Iron gall ink is produced by treating a solution of tannins with iron(II) sulfate [82].

Tannins can also be used as a mordant, and are especially useful in natural dyeing of cellulose fibers such as cotton. The type of tannin used may or may not have an impact on the final color of the fiber [83].

2.6.7. Terpenoids

Terpenoids are a large and diverse class of naturally occurring organic chemicals derived from the 5-carbon compound isoprene, and the isoprene polymers called terpenes. They are sometimes called isoprenoids. Most are multi-cyclic structures with oxygen-containing functional groups. About 60% of known natural products are terpenoids [84]. Although sometimes used interchangeably with "terpenes", Terpenoids contain additional functional groups usually O-containing Terpenes are hydrocarbons [85].



Scheme 2. 3. Chemical structure of terpenoid taxol an anticancer drug.

Plant terpenoids are used for their aromatic qualities and play a role in traditional herbal remedies. It uses prevention and treatment of cancer, anti-microbial, anti-fungal, anti-viral, anti-hyperglycemic, anti-inflammatory, anti-oxidant, and anti-parasitic and also skin permeation enhancer [86].

2.7. Preliminary Phytochemical Screening

2.7.1. Phytochemical studies

Phytochemical studies of plants, animals and microorganisms help to have deeper understanding of the factor underlying for the growth, development and differentiation of their chemistry out comes.⁸⁷ Plants and animals produce very large numbers of chemicals. These chemical products are classified into primary and secondary metabolites. Primary metabolites are those, which are common to all species and subdivided into proteins, carbohydrates, lipids and nucleic acids. Primary metabolites are essentially ubiquitous and certainly essential for life and they function in cycle [88]. The secondary metabolite are often referred to as “natural products” secondary metabolites can be subdivided in to alkaloids, flavonoid, glycosides, saponins, steroids, tannins, terpenoids, etc. The classification were based on the means by the materials were they made. In opposite, secondary metabolites are non- essential to life but they definitely contribute to the species fitness of survival. Different plant and animal species can employ different biosynthetic route to produce the same metabolites [89]. The amount of plant-derived drugs being used demonstrated by the contribution of natural products to the development of medicine. Many backgrounds demonstrate that plant growth-regulating hormones, fungicides and agricultural chemicals developed from natural products from time to time [90].

2.7.2. Phytochemical screening

Phytochemicals are the plant derived chemicals and have the capability of disease prevention, thus beneficial to human health. Phytochemicals produced by plants are two types; Primary metabolites and Secondary metabolites. Since ancient times and nowadays secondary metabolites, being an important source of drug and almost 50% of the practical drugs used were

derived from natural source. Secondary metabolites like alkaloids, tannins, phenols, flavonoids, saponins, antraquinones, cardiac glycosides and cyanogenic glycosides etc., have pivotal importance. Plant derived compounds are highly varied in structure many are aromatic substances, most of which are phenols or their oxygen-substituted derivatives [91].

2.8. The role of *E.abbyssinica* Species and its Antimicrobial activity

The traditional medicinal plants still play an important role in protecting serious diseases and to achieve the basic health needs commonly in the unindustrialized countries. Recently antibiotics study on plants increases progressively to solve therapeutic problems of the world [92]. Plants are mainly used as a source of food additive and their extracts also own antibacterial, antifungal, antiviral and antioxidant assets [93]. The most active phytochemicals or secondary metabolites are antibiotic principle of plants that offer protection against diseases, boosting the immune system and inhibit growth of bacteria [94]. Studies illustrate that latex extracted plants have antibacterial and antifungal activity against below the table [95]. It was performed by agar disc diffusion and agar well diffusion method [96].

Table 2. Codes and types of pathogen.

No	Pathogen name	Abbreviation	Type	Code
1	Candida alibi cans	Ca	Yeast	Extracted from patient
2	Escherichia coli	Ec	Gram- bacteria	ATCC 25922
3	Listeria mono-cytogenes	Lm	Gram+ bacteria	ATCC 19115
4	Pseudomonas aeruginosa	Pa	Gram- bacteria	ATCC 27853
5	Staphylococcus aureus	Sa	Gram+ bacteria	ATCC 25923
6	Salmonella typhimurium	Sa1	Gram- bacteria	ATCC13311
7	Staphylococcus epidermidis	Se	Gram+ bacteria	ATCC12228
8	Shigella sonnei	Sh	Gram- bacteria	ATCC25931
9	Streptococcus pyogenes	Sp	Gram+ bacteria	ATCC19615

CHAPTER THREE

3. MATERIALS AND METHODS

3.1. Plant Materials

3.1.1. Identification of euphorbia abyssinica species for the study

The species of *E.abyssinica* used in this study were identified by their shoot at agricultural research center in Burie area, and collected from Godber Kebele Dembecha Woreda in west Gojjam zone, Ethiopia, which is located about 350 Km away from the capital city, Addis Abeba.

3.1.2. Collection of the part of *E.abyssinica*

The latex part of *E.abyssinica* was punctured or perforation by small fax and collected by the tup with a lid that was made from a hose from the *E.abyssinica* tree in august, 2020 from the crop of pale in Godber Kebele. Then packed by polystyrene, because the latex do not percolate.

3.2. Chemicals and Equipment

3.2.1. Chemicals

Note: All extraction solvents with manufactured in 2017 and expired in 2022 have analytical grade and purchased from ZAF pharmaceutical PLC in Addis Ababa city.

- ❖ Chemicals like petroleum ether, ethyl acetate, and ethanol were used as an organic solvent for the extraction of bioactive components from the latex part of the plant.
- ❖ General laboratory reagents like concentrated hydrochloric acid, 5% ferric chloride solution (FeCl_3), concentrated sulfuric acid, 2N sodium hydroxide, Chloroform (CHCl_3), 10% ammonium (NH_3) solution, Acetic anhydride, Distilled water and Mayer's reagent

(Potassium mercuric iodide) were used in this experiment from Debre Berhan university organic laboratory for phytochemical test.

- ❖ Appropriate media for bacteria (Muller hinter agar) was used to develop microorganisms in Petri-dishes. Chloramphenicol 0.02 g/ml was used as positive control to test anti-bacterial activity and DMSO used as a negative control and to dilute the crude extracts in a biology laboratory of Debre Berhan University.

3.2.2. Materials and equipment

Materials and types of equipment that used during this study were

- Spatula to take sample into electronic analytical balance.
- Beam balance or Electronic analytical balance to measure the required amount of sample.
- Erlenmeyer flasks and conical flask were used to soak the sample with the solvent measured by measuring cylinder.
- Glass rod or stirrer was used to mix the sample with the solvent.
- Complete Soxhlet extractor apparatus:
 - i Heating mantel used for providing necessary heat to the solvent which is kept in round bottom flask.
 - ii Round bottom flask used to collect the extracted sample.
 - iii Few glass pieces to avoid bumping in side flasks.
 - iv Extraction assembly with Drug chamber consists of vapor outlet, crude drug chamber and solvent inlet.
 - v The sample filled in closed membrane which is called Thimble, can made by using Filter paper or muslin cloth, and used to prevent leakage of sample powder which can damage The Soxhlet extraction assembly.
 - vi Condenser convert vapor in to liquid.
 - vii Clamp used to giving support and holding the various assembly Soxhlet extractor.
- Dropper, test tubes and different sizes beakers used to conduct phytochemical test.
- Vials were used to hold the extracts.

- Aluminum foil used to cover the extracted sample
- Rotary evaporator was used to remove solvents from the crude extract
- Petri-dishes were used to hold inoculated nutrient agar for bacterial growth, 6mm diameter Whatman No 4 paper discs, forceps, Autoclave; boiling water bath for melting the solid media, incubator and holder were used.
- Bunsen burner, micropipette, micropipette tips, Laminar air flow hood, spreader, inoculating loop
- Elements for personal biosecurity (gloves, gowns, goggles or protective eye wear, chemical/biological safety hood) were the apparatus and instruments used in this experiment.

3.3. Solvent Extraction Procedure of the Plant Material

45 gm. of fresh latex was soaked in thimble and put in Soxhle chamber, 200ml of petroleum ether transferred into round bottom flask and put in heating mantle then setup Soxhlet extractor after then on the heating mantle at temperature of 75°C for 18 hours. After 18 hours take out the round bottom flask from the heating mantle and transferred into Erlenmeyer flask then put into refrigerator for 6 hours then after filtered by filter paper and evaporate by rotatory evaporator. Finally, the crude extracts were dried and weighed.

For the same procedure measure 200 ml of ethyl acetate and transferred into round bottom flask for the Soxhlet extractor for the same temperature stayed for 24 hours. After 24 hours take out round bottom flask and transferred in to Erlenmeyer flask then put into refrigerator for 6 hours then after filtered by filter paper and evaporated by rotatory evaporator. Finally, the crude extracts were dried and weighed.

For the same procedure measured 200ml of ethanol transferred to round bottom flask stayed for 24 hours.

After collecting the extract it is stored in a well-closed container and covered with aluminum foil and stored in the refrigerator. The extract is diluted with the solvent and administered to carry out further phytochemical analysis and antibacterial tests.

3.4. Qualitative Phytochemical Screening:

3.4.1. Test for Alkaloids:

2 ml of extract was taken by test tube, 2 ml of concentrated hydrochloric acid was added. Then few drops of Mayer's reagent were added. Presence of green color or white precipitate indicates the presence of alkaloids.

3.4.2. Test for Flavonoids:

2 ml of extract was taken by test tube, 2 ml of 2N sodium hydroxide was added. The presence of yellow color indicates the presence of flavonoids.

3.4.3. Test for Glycosides:

2 ml of extract was taken by test tube, 3ml of chloroform and 10% ammonia solution was added. The formation of pink color indicates the presence of glycosides.

3.4.4. Test for Saponins:

2 ml of extract was taken by test tube, 2 ml of distilled water were added and shaken for 15 min lengthwise. It resulted in the formation of 1 cm layer of foam that indicated the presence of saponins

3.4.5. Test for Steroids:

2 ml of extract was taken by test tube, 2 ml of acetic anhydride and 2 ml of concentrated Sulphuric acid added it resulted in change violent to blue or green color indicates the presence of steroids.

3.4.6. Test for Tannins:

1 ml of extract was taken by test tube, 2 ml of 5% ferric chloride was added. The formation of dark-blue or greenish-black indicates the presence of tannins.

3.4.7. Test for Terpenoids:

0.5 ml of the extract was treated with 2 ml of chloroform and concentrated Sulphuric acid. The formation of red-brown color at the interface indicates the presence of terpenoids.

3.5. Test of Antimicrobial Activity of the Crude Extracts

3.5.1. Preparation of microbial

Inoculum of *Candida albicans* yeast and *Escherichia coli*, *Listeria monocytogenes*, *Pseudomonas aeruginosa*, *Staphylococcus aureus*, *Salmonella typhimurium*, *Staphylococcus epidermidis*, *Shigella sonnei*, and *Streptococcus pyogenes* bacterial were prepared in isolated nineteen (18) Petri-dishes containing nutrient agar. 200 ml aliquots of the inoculated nutrient agar were distributed in sterile separate Petri-dishes. 2 mL of a standardized organism stock suspension was thoroughly mixed with 200 mL of a sterile molten nutrient agar and left to set in incubator at 37°C for 24 hours the incubation of bacteria.

3.5.2. Antibacterial activity test

The paper-disc agar diffusion method was adopted to assess the antibacterial activity of the DMSO diluted petroleum ether, ethyl acetate and ethanol extracts of the latex of *E. abyssinica* and fresh latex. This was performed against gram positive bacteria (Lm, Sa, Se, Sp), gram negative (Ec, Pa, Sal, Sh) bacterial and Ca yeast. Gram positive bacteria give a positive result in a gram stain test and gram negative bacteria do not show a positive result in gram staining test. The paper disks which are prepared from the Whatmann filter paper Number 4 (6mm in diameter) were sterilized in autoclave for 1 hour and placed individually with sterile forceps, then gently pressed down onto the agar disc. 20 µl of the concentrations (100, 200 and 250 mg/ml) of each sample were pipetted to the discs in four replications using microliter pipette. The extracts were allowed to diffuse on the incubated bacteria at room temperature for 24 hours. Four replicates were carried out for each extract against each of the bacterial organisms in separate Petri dishes. A cup was made for each organism. After incubation, the diameters of the growth inhibition zones were measured in millimeters and compared to Chloramphenicol 0.02g/ml standard and the average values were tabulated.

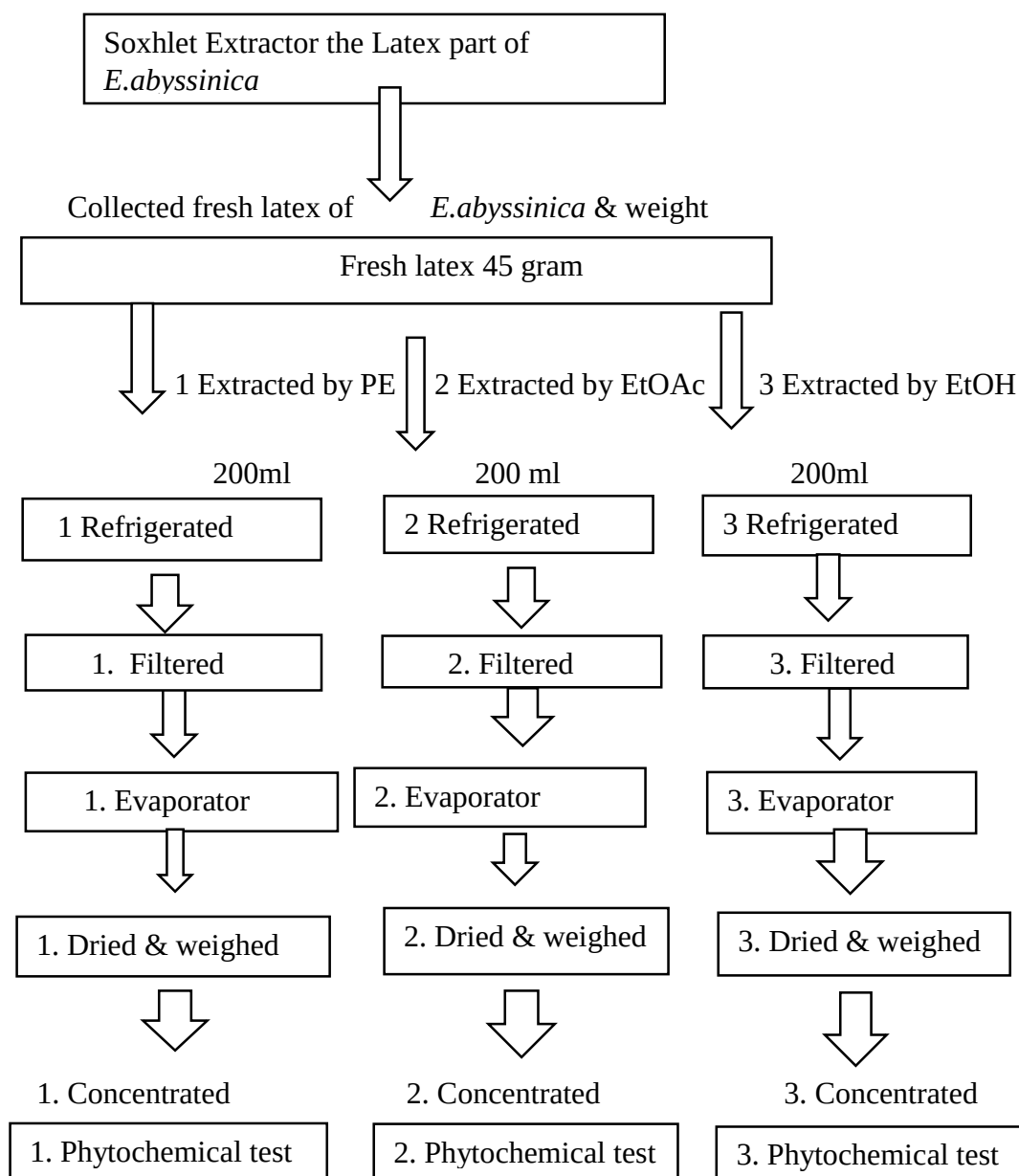


Figure 2. Schematic diagram for a sample extraction and isolation extracts of *E.abyssinica*.

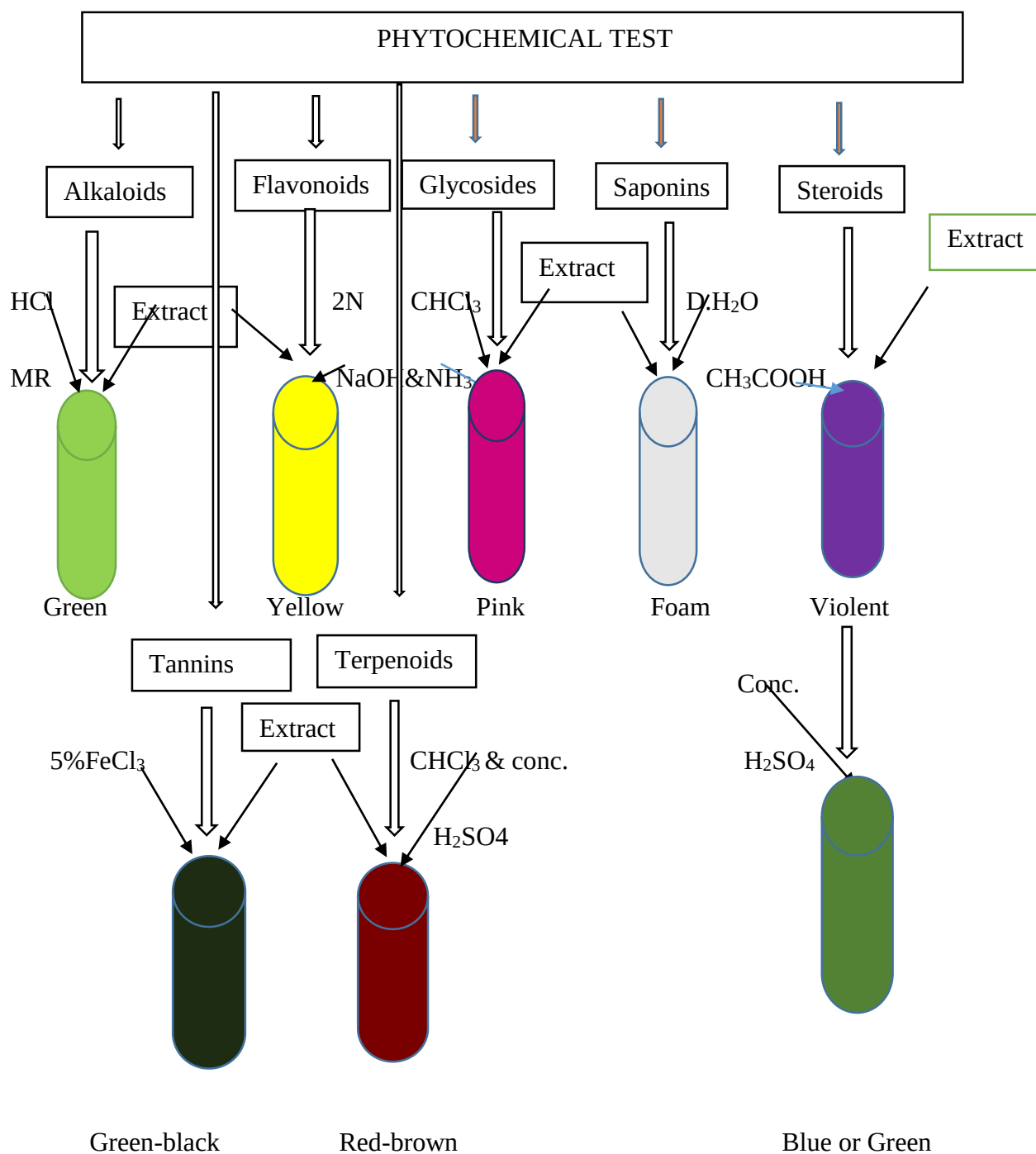


Figure 3. Schematic diagram for phytochemical test of the extracted latex sample of *E. abyssinica*.

CHAPTER FOUR

4. RESULT AND DISCUSSION

4.1. Sample extraction

The yields of solvent extracts (by Soxhlet extractor of the latex part of *Euphorbia abyssinica*) using different organic solvents were given in Table 3. The crude extracts separated by different organic solvents and concentrated by rotary evaporator give different yield. This shows the nature of solvents to dissolve ingredients found in the sample was different. The percentage yield of plant sample and extraction methods were shown in Table 3.

Table 3. The percentage yield of sample and extraction method.

Plant sample (g)	Solvent / volume	Extracted (g)	Extracted yield	Method of extraction
45 g sample	Petroleum ether 200ml	1.53	3.40 %	Soxhlet extractor
45 g sample	Ethyl acetate 200ml	3.14	6.98 %	Soxhlet extractor
45 g sample	Ethanol 200ml	1.43	3.18 %	Soxhlet extractor

4.2. Phytochemical Constituents of the latex part of *E.abysynica* Extracts

The crude extracts of PE, EtOAc and EtOH of the latex of *E.abysynica*, were subjected to qualitative phytochemical tests to identify various classes of bioactive chemical constituents such as alkaloid, flavonoids, glycosides, saponins, steroids, tannins, and terpenoids using appropriate reagents and following standard methods. Phytochemical test of petroleum ether extracts revealed the presence of terpenoids and saponins, while flavonoids, steroids and tannins were found moderately. Phytochemical test of the ethyl acetate extracts revealed the presence flavonoids, saponins and terpenoids whereas alkaloids, glycosides, steroids and tannins were found moderately. Flavonoids and terpenoids were found in the EtOH extract and

also steroids and tannin were found moderately for this extract. Preliminary phytochemical analysis show the presence of major classes of secondary metabolites in ethyl acetate extract as displayed in Table 4.

Table 4. Phytochemical constituents of different solvent extracted of the latex part of *E.abysinnica*.

No	Phytochemical parameters	Observed color (ppt)	Test	The latex Extracts		
				PE	EtOAc	EtOH
1	Alkaloids	Green color or white ppt	Mayer's	—	+	—
2	Flavonoids	Yellow	Sodium hydroxide	+	++	++
3	Glycosides	Pink	Borntranger's	—	+	—
4	Saponins	Layer of foam	Foam	++	++	—
5	Steroids	Blue or green	Liebermann-Bur chards	+	+	+
6	Tannins	Dark-blue or Green-black	Ferric chloride	+	+	+
7	Terpenoides	Red-brown		++	++	++

Key: ++ present in more quantity, + present moderately and _ absent

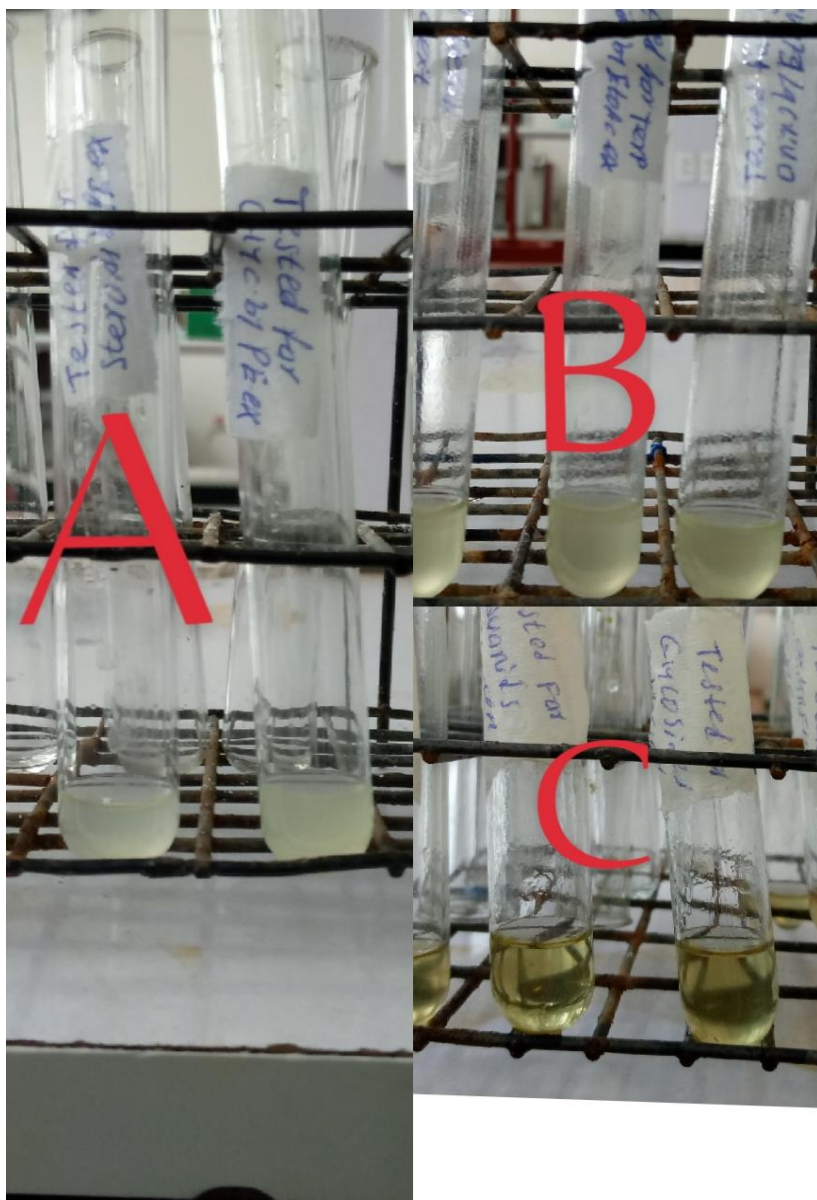


Figure 4. Phytochemical test references.

A] Petroleum ether extracted B] Ethyl acetate extracted and C] Ethanol extracted.

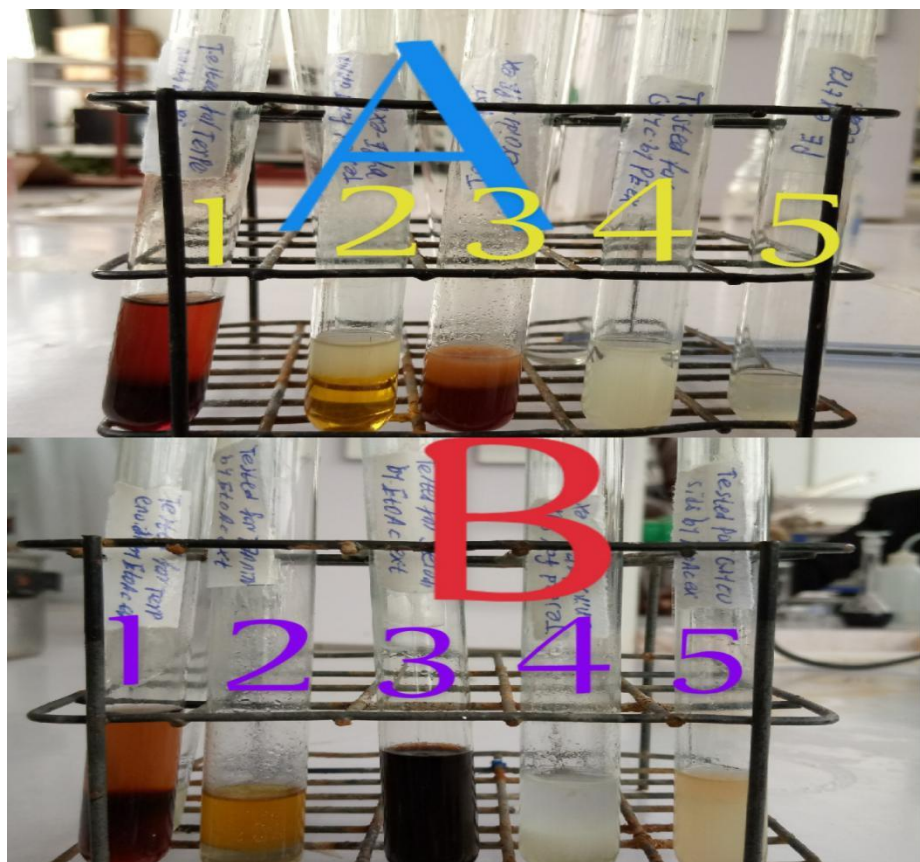


Figure 5. Result of phytochemical test extracted by petroleum ether and ethyl acetate.

A] Petroleum ether extracted:

1. Terpenoids: It is red brown color, indicates the latex have terpenoids.
2. Tannins: Approximately it is greenish-black color, the latex have moderately tannins.
3. Steroids: It also approximately blue-green color, the latex have steroids.
4. Glycosides: It is a white color i.e. no have glycosides.
5. Alkaloids: have same to 4.

B] Ethyl acetate extracted:

1. Terpenoids, 2. Tannins, and 3. Steroids, have the same to with A.
4. Saponins, and 5. Glycosides: It is approximately a pink color i.e. have moderately glycosides.

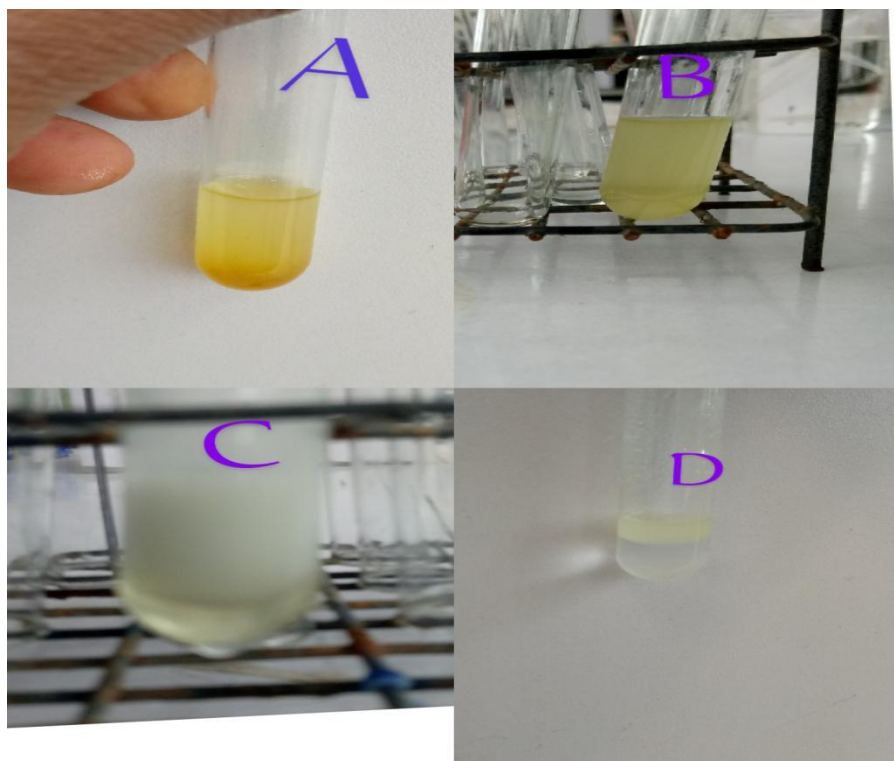


Figure 6. Result of other phytochemical test by petroleum ether and ethyl acetate extracted.

A]. Flavonoids by EtOAc extracted. B]. Flavonoids by PE extracted.

C]. Saponins by EtOAc extracted. And D]. Saponins by PE extracted.

C and D are shown after shaken 15 minutes.

The color of A is yellow i.e. the latex have Flavonoids in EtOAc extracted. And B is approximately a yellow color i.e. the sample have moderately Flavonoids in PE extracted. Both C and D are foam color in 1 cm layer indicates the latex have Saponins in EtOAc and PE extracted.

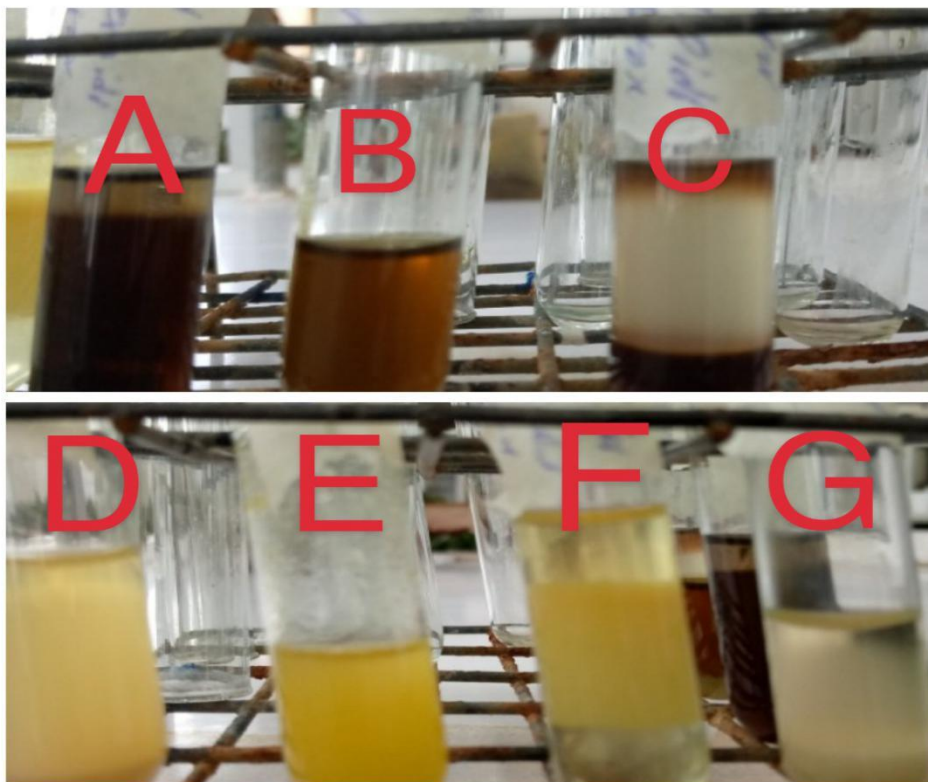


Figure 7. Result of phytochemical test extracted by ethanol.

A].Steroids, B]. Tannins, C]. Terpenoids, D]. Alkaloids, E]. Flavonoids, F]. Glycosides, and G], Saponins.

The latex have tannins and flavonoids in ethanol extracted, because tannins have Greenish-black color and Flavonoids have yellow color. Whereas Steroids and terpenoids were found moderetley in EtOH extracted.

4.3. Anti-Bacterial Activity Test

The PE, EtOAc and EtOH extracts of latex *E.abbyssinica* and fresh latex were subjected to anti-bacterial test on gram positive:- *Listeria mono-cytogenes*, *Staphylococcus aureus*, *Staphylococcus epidermidis*, *Streptococcus pyogenes* and gram negative:- *Escherichia coli*, *Pseudomonas aeruginosa*, *Salmonella typhimurium*, *Shigella sonnei* and also Yeast:- *Candida alibi cans*. The results were investigated by comparing with a positive control drug Chloramphenicol 0.02g/ml Zone of growth inhibition of latex *E.abbyssinica* extracts were measured using ruler and results are tabulated by taking the mean average results of the three trials. The latex extracts and fresh latex of *E.abbyssinica* shows antibacterial activity against on *Escherichia coli*, *Staphylococcus epidermidis*, *Shigella sonnei* bacterial and *Candida alibi cans* yeast. The petroleum ether extracts of the sample show on Ec and Sh bacterial with inhibition zone 10 mm and 9 mm were observed respectively. The ethyl acetate extracts of the sample show on Ec, Sh, bacterial and Ca yeast with inhibition zone ranges from 6 mm to 8 mm, 6 mm to 7 mm and 7 mm to 10 mm were observed respectively. And the ethanol extracts of the sample show on Ec, Se, Sh, bacterial and Ca yeast with inhibition zone ranges from 7 mm to 10 mm, 8 mm to 9 mm, 7 mm to 8 mm and 7 mm to 10 mm were observed respectively. And also the fresh latex of the sample show on Ec and Sh bacterial with inhibition zone 9 mm and 7 mm. The maximum antimicrobial activities were observed at concentration of 250 mg/ml of the extract. The inhibition zone decreases as the concentration decreases. The ethanol extract was found effective against compared to the other extracts. The other bacterial strains had no effective against by sample extracts as shown in Table 4.3.

Table 5. Anti-bacterial activity of crude extract resulted from disc-diffusion assay.

No	Microorganisms	Concentration mg/ml	Zone of inhibition in mm				
			PE	EtOAc	EtOH	F La	Drug
1	Ca yeast	100	–	–	7	–	25
		200	–	7	8	–	
		250	–	10	10	–	
2	Ec bacteria	100	–	–	7	–	12
		200	–	6	8	6	
		250	10	8	10	9	
3	Lm bacteria	100	–	–	–	–	28
		200	–	–	–	–	
		250	–	–	–	–	
4	Pa bacteria	100	–	–	–	–	10
		200	–	–	–	–	
		250	–	–	–	–	
5	Sa bacteria	100	–	–	–	–	25
		200	–	–	–	–	
		250	–	–	–	–	
6	Sa1	100	–	–	–	–	10
		200	–	–	–	–	
		250	–	–	–	–	
7	Se	100	–	–	–	–	25
		200	–	–	8	–	
		250	–	–	9	–	
8	Sh	100	–	6	–	–	20
		200	–	7	7	–	
		250	9	7	8	7	
9	Sp	100	–	–	–	–	25
		200	–	–	–	–	

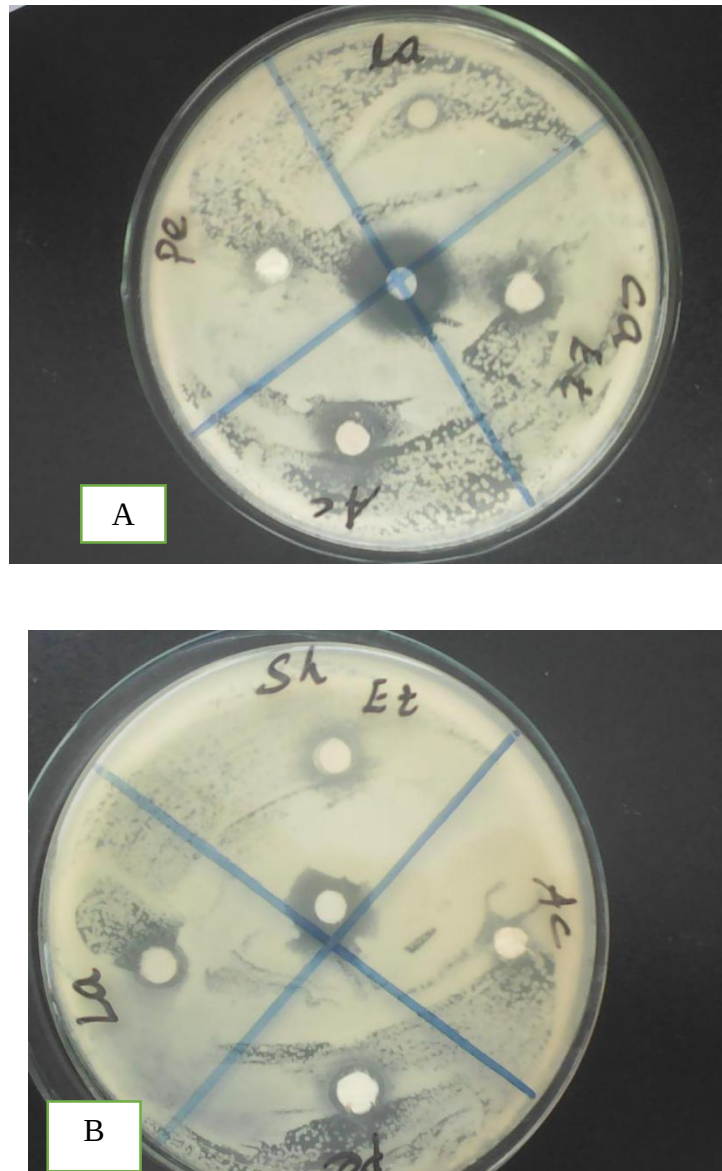


Figure 8. Ca yeast and Sh bacteria test and inhibition zone observed by latex extracted and fresh latex.

A]. *Candida albicans* yeast. B]. *Shigella Sonnei* Bacteria

Key [The code of mine]: Ac- Ethyl acetate, Et- Ethanol, La- Fresh latex, and Pe- Petroleum ether.

CHAPTER FIVE

CONCLUSIONS AND RECOMMENDATIONS

5.1. Conclusions

Based on the observation and investigations provided in the thesis, the preliminary phytochemical investigation of ethyl acetate solvent extract of *latex E.abyssinica* showed the presence of major phytochemicals. But all solvent (PE, EtOAc and EtOH) extracts show the existence of phytochemicals. These are flavonoids, Steroid, saponin, terpenoids and Tannins. The latex extract and fresh latex of *E.abyssinica* shows antibacterial activity against on *Escherichia coli*, *Staphylococcus epidermidis*, *Shigella sonni*, bacterial and candida alibi cans yeast. From these test the ethanol extract was found effective against compared to the other extracts.

5.2. Recommendation

The latex part of *Euphorbia abyssinica* contains different phytochemicals (flavonoids, saponins, steroids, tannins and terpenoids). From the phytochemical investigation one may be responsible for the prevention and treatment of cancer, anti-microbial, anti-fungal, anti-viral, anti-hyperglycemic, anti-inflammatory, anti-oxidant, and anti-parasitic and also skin permeation enhancer properties of this herbal plant.

So it is recommended that the plant has anti-bacterial importance. Further studies are needed to undertake its chemical property, bioactivity and toxicity profile and also emphasis on the mechanisms of action of bioactive constituents of the species to illustrate the correlation between the ethno medicinal uses and pharmacological properties of the species. Further studies also determine the characterization of the compound by using IR and GC-MS instruments. Also determines the total phenolic content and DPPH radical scavenging activity it is important to deal the detailed toxicological review of the crude extracts and pure compounds of the species.

6. REFERENCE

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